THE MASTERMIND COACHING CLASSES

Revision Notes

Class 8 · Mathematics

Ganita Prakash — Part 2, Chapters 8 to 14 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
8	Algebra Play	Algebra tricks: think-of-a-number, number pyramids, grids, digit puzzles, divisibility, word problems
9	Area	Area of rectangles, triangles, polygons, parallelograms, rhombuses, trapeziums; units of area
10	Exploring Some Geometric Themes	Fractals (Sierpinski carpet and gasket, Koch snowflake), solids and nets, shortest paths, projections, isometric drawings
11	Fractions in Disguise	Percentages, increase and decrease, profit and loss, discount, tax, simple interest vs compounding, depreciation
12	Proportional Reasoning – 2	Ratios with many terms, dividing a whole, pie charts, direct and inverse proportion
13	Tales by Dots and Lines	Mean as a balance point, effect of changes on mean and median, frequency tables, spreadsheets, line graphs
14	The Baudhāyana–Pythagoras Theorem	Baudhāyana–Pythagoras theorem, √2, Baudhāyana triples and applications

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CHAPTER 08 · MATHS (PART 2)

Algebra Play

Algebra is the tool that explains why number tricks always work. In this chapter we turn tricks, pyramids, calendar grids, digit puzzles and divisibility rules into expressions and equations, solve them, and use the same idea to invent our own puzzles.

Big Questions

- How does algebra explain ‘think of a number’ tricks?
- How can letter-numbers justify patterns in pyramids, grids and divisibility?
- How do we form and solve equations for word problems?

‘Think of a Number’ Tricks

Call the starting number **x** and write each instruction as an expression. If the letter cancels at the end, the answer is the same for every starting number.

Instruction	Expression
Think of a number	x
Multiply by 4	4x
Add 12	4x + 12
Divide by 4	x + 3
Subtract the number you started with	x + 3 − x = **3**

IMPORTANT

The trick works because **x cancels**. To make the final answer another value, change the numbers added, for example add 20 in place of 12 to end with 5.

▸ Guessing a date

Let the month be **M** and the day **D**. Follow: multiply M by 5, add 6, multiply by 4, add 9, multiply by 5, add D.

Step	Expression
× 5	5M
+ 6	5M + 6
× 4	20M + 24
+ 9	20M + 33
× 5	100M + 165
+ D	100M + 165 + D

FORMULA

Final answer = **100M + 165 + D**. Subtract **165**; the last two digits are the day D (at most 31) and the digits before them are the month M.

- Date 7 March (M = 3, D = 7): the steps give 15 → 21 → 84 → 93 → 465 → **472**.
- If the answer is **1177**: 1177 − 165 = 1012, so M = 10 and D = 12, i.e. 12 October.
- Different steps need a different number to subtract; work it out with M and D first.

Number Pyramids

In a number pyramid each number is the **sum of the two numbers directly below it**.

- To fill a pyramid with missing numbers, give letters to unknown boxes and write one equation for each rule.
- Use subtraction to move up or down: if the top is 10 and one box below is 4, the other is 10 − 4 = 6.

▸ Worked example — missing middle number

Top = 52, second row = a, b, bottom row = 10, c, 4.

Step 1 — Write rules
a = 10 + c, b = c + 4 and a + b = 52.

Step 2 — Combine
(10 + c) + (c + 4) = 52, so 14 + 2c = 52.

Step 3 — Solve
2c = 38, so **c = 19**.

Step 4 — Back-fill
a = 29 and b = 23. Check: 29 + 23 = 52.

FORMULA

Bottom row a, b, c (3 rows): top = **a + 2b + c**. Bottom row a, b, c, d (4 rows): top = **a + 3b + 3c + d**. The coefficients are 1, 2, 1 and 1, 3, 3, 1.

- Bottom row 3, 5, 2: top = 3 + 10 + 2 = **15** (rows above: 8, 7, then 15).
- Bottom row 2, 1, 4, 3: top = 2 + 3 + 12 + 3 = **20**.

Fun with Grids

▸ Calendar grid

Take a 2 × 2 block of a calendar. If the top-left number is **a**, the block is a, a + 1, a + 7, a + 8.

FORMULA

Sum of a 2 × 2 block = a + (a + 1) + (a + 7) + (a + 8) = **4a + 16**.

- If the sum is 52: 4a + 16 = 52, so 4a = 36 and **a = 9**; the block is 9, 10, 16, 17.
- For a 3 × 3 block with centre **c**, the sum is **9c** (the numbers on opposite sides of c add to 2c in pairs).
- Because a week has 7 days, the number below a date is the date **+ 7**.

▸ Algebra grids

In a grid where each row's last entry is the sum of the shapes before it, treat each shape as a letter-number, find the repeated shape first, then substitute.

The Largest Product

Fill the digits 2, 4, 7 in the boxes of (two-digit number) × (one-digit number), using each once. There are 6 arrangements; group those with the same multiplier and keep the one whose tens digit is larger.

> **IMPORTANT**
> For digits p < q < r: the largest product is **(qp) × r**, i.e. the **largest digit is the multiplier** and the other two digits form the multiplicand in **decreasing** order.

- Reason: qp × r = 10qr + pr and rp × q = 10rq + pq; the first parts are equal and pr > pq because r > q.
- Digits 2, 4, 7: the largest product is 42 × 7 = **294** (compare 72 × 4 = 288, 24 × 7 = 168).

Decoding Divisibility Tricks

Trick	Algebra	Result
Reverse a 2-digit number, subtract (ba − ab)	(10b + a) − (10a + b) = 9(b − a)	Divisible by **9**; quotient is b − a (83 → 38: difference 45 = 9 × 5)
Reverse a 2-digit number, add	(10a + b) + (10b + a) = 11(a + b)	Divisible by **11** (47 + 74 = 121 = 11 × 11)
Cycle a 3-digit number, add abc + bca + cab	111a + 111b + 111c = 111(a + b + c) = 37 × 3 × (a + b + c)	Divisible by **37** and by 3 (123 + 231 + 312 = 666 = 37 × 18)
Repeat a 3-digit number: abcabc	abc × 1001 = abc × 7 × 11 × 13	Dividing by 7, 11, 13 in turn gives back **abc**

> **NOTE**
> Writing a 2-digit number as **10a + b** (not a × b) and a 3-digit number as **100a + 10b + c** is the key step in every digit trick.

Word Problems with Letter-Numbers

Step 1 — Choose a letter
Let the unknown quantity be x.

Step 2 — Translate
Write each sentence as an expression.

Step 3 — Form the equation
Join two expressions for the same quantity with =.

Step 4 — Solve and check
Solve, then test the answer in the original story.

Problem	Equation	Answer
Heads 40, legs 110 (horses and hens)	If h horses: 4h + 2(40 − h) = 110	15 horses, 25 hens

Problem	Equation	Answer
Mother is 4 times daughter's age; in 5 years she will be 3 times	4x + 5 = 3(x + 5)	Daughter 10, mother 40
Two friends: B has twice as many cows as A; after B gives 3 to A they are equal	2x − 3 = x + 3	A has 6, B has 12
Cart rent ₹4000, cost ₹8 per dosa, 200 dosas, profit ₹1600	200p − 4000 − 8 × 200 = 1600	Price p = ₹36

> **NOTE**
> Without letters, the horses–hens problem can also be solved: if all 40 animals were hens there would be 80 legs; the extra 30 legs come from horses (2 extra each), so there are 15 horses.

▸ Doubling and paying – the genie problem

A person has x coins; each round the coins double and 16 are paid. After 3 rounds: round 1: 2x − 16; round 2: 4x − 48; round 3: 8x − 112.

- If nothing is left, 8x − 112 = 0, so **x = 14** (14 → 12 → 8 → 0).
- The deal helps the person only if the cost per round is **less than the coins held** at that round.
- Sum of the first n odd numbers is n^2, so (1 + 3 + 5) ÷ (7 + 9 + 11) = 9 ÷ 27 = 1 ÷ 3.

Common Mistakes

Mistake	Correct
2(x + 3) = 2x + 3	2(x + 3) = 2x + 6
Writing the number ab as a × b	ab = 10a + b
Forgetting to subtract the same number from both sides	Do the same operation on both sides
Not checking the answer in the story	Substitute back and test

Likely question	Model answer
Why does a 'think of a number' trick give a fixed answer?	The letter cancels in the last step, so the result does not depend on the starting number.
In the date trick, what is subtracted from the final answer?	165 (with the steps ×5, +6, ×4, +9, ×5, +day).
What is the top number of a 3-row pyramid with bottom a, b, c?	a + 2b + c.
Why is the difference of a 2-digit number and its reverse divisible by 9?	It equals 9(b − a).
How do we form the largest product with three digits?	Take the largest digit as the multiplier and arrange the other two in decreasing order as the multiplicand.

★ KEY TAKEAWAYS

- Write each instruction as an expression; if the letter cancels, the trick always works.
- Date trick: answer = 100M + 165 + D, so subtract 165 to read M and D.
- 3-row pyramid top = a + 2b + c; 4-row top = a + 3b + 3c + d.
- 2 × 2 calendar block sum = 4a + 16; 3 × 3 block sum = 9 × centre.
- Largest product: biggest digit as multiplier, other digits in decreasing order.
- ba − ab = 9(b − a); ab + ba = 11(a + b); abc + bca + cab = 111(a + b + c); abcabc = abc × 7 × 11 × 13.
- Form equations from the story, solve, then check in the original problem.

CHAPTER 09 · MATHS (PART 2)

Area

Area measures how many unit squares fit inside a region. Starting from the rectangle, this chapter builds the area of a triangle, then of any polygon, a parallelogram, a rhombus and a trapezium by cutting and rearranging shapes, and ends with real-life units of area.

Big Questions

- Why is perimeter not a measure of area?
- How do we get the area of a triangle, and from it the area of any polygon?
- How are the areas of a parallelogram, rhombus and trapezium found by dissection?

Area of a Rectangle and Why Perimeter Fails

FORMULA

Area of a rectangle = **length × width**. A 9 cm × 6 cm rectangle holds 9 × 6 = **54 unit squares**, so its area is 54 cm².

- The diagonal of a rectangle splits it into **two congruent triangles**, so each has half the area: ½ × 9 × 6 = **27 cm²**.
- Area is measured in **square units** (cm², m²). Perimeter is a length in cm or m, so it cannot measure area.

Rectangle	Perimeter	Area
10 cm × 2 cm	24 cm	20 cm²
6 cm × 6 cm	24 cm	36 cm²
12 cm × 1 cm	26 cm	12 cm²
5 cm × 5 cm	20 cm	25 cm²

IMPORTANT

Same perimeter can give different areas (rows 1 and 2). A **larger perimeter can even give a smaller area** (rows 3 and 4). So perimeter is not a measure of area.

▸ Path around a rectangle

Outer rectangle 20 m × 14 m, inner park 16 m × 10 m (path 2 m wide all round).

FORMULA

Area of the path = area of outer rectangle − area of inner rectangle = 280 − 160 = **120 m²**.

Area of a Triangle

FORMULA

Area of a triangle = **½ × base × height**, where the height is the perpendicular distance from the opposite vertex to the base (or the base extended).

- Any triangle can be placed inside a rectangle with the same base and height; it fills exactly **half** of it.
- For an obtuse triangle, subtract: area = ½ × h × DC − ½ × h × DB = ½ × h × BC.
- Base 12 cm, height 7 cm: area = ½ × 12 × 7 = **42 cm²**.

▸ Finding an altitude

In △ABC take BC = 10 cm with altitude AX = 6 cm, and AC = 8 cm.

Step 1 — Area using BC
½ × 6 × 10 = 30 cm².

Step 2 — Area using AC
½ × BY × 8 = 4 × BY, where BY is the altitude from B to AC.

Step 3 — Equate
4 × BY = 30, so **BY = 7.5 cm**.

IMPORTANT

A line joining a vertex to the **midpoint of the opposite side** divides the triangle into two triangles of **equal area** (same height, equal bases). So the four triangles formed by the diagonals of a rectangle have equal areas.

- All triangles on the **same base** with the third vertex on a line **parallel to the base** have the **same area**.
- Among them the one with the **least perimeter** has its vertex on the perpendicular bisector of the base (found by reflecting in the parallel line).

Area of Any Polygon

Every polygon can be divided into triangles. Find the area of each triangle and add.

Step 1
Join vertices with diagonals so that the polygon splits into triangles.

Step 2
Measure a base and its perpendicular height for each triangle.

Step 3
Add the areas of the triangles.

- Quadrilateral ABCD with diagonal AC = 22 cm and perpendiculars 3 cm and 3 cm from B and D to AC: area = ½ × 22 × 3 + ½ × 22 × 3 = ½ × 22 × (3 + 3) = **66 cm²**.
- Diagonal 18 cm with perpendiculars 5 cm and 7 cm: area = ½ × 18 × (5 + 7) = **108 cm²**.

FORMULA

Quadrilateral = **½ × diagonal × (sum of the two perpendiculars to it)**.

Parallelogram

Cut a right-angled triangle from one end of the parallelogram and attach it at the other end; the result is a **rectangle** of the same area (base × height).

FORMULA

Area of a parallelogram = **base × height** (height is perpendicular to the base, not the slant side).

- Base 15 cm, height 8 cm: area = **120 cm²**.

- If the other side is 10 cm, the height on that side is 120 ÷ 10 = **12 cm**: any side can be the base with its own height.
- A diagonal divides a parallelogram into two triangles of equal area.

Rhombus

FORMULA

Area of a rhombus = **½ × d_1 × d_2** (product of the diagonals); a rhombus is also a parallelogram, so base × height works.

- Diagonals 16 cm and 12 cm: area = ½ × 16 × 12 = **96 cm²**.
- The diagonals bisect each other at right angles, so each side = √(8^2 + 6^2) = **10 cm**.
- Check by triangles: △ADB and △CDB each have area ½ × AO × BD and ½ × CO × BD; their sum is ½ × AC × BD.

Trapezium

FORMULA

Area of a trapezium = **½ × height × (sum of the parallel sides)**.

- Parallel sides 9 cm and 15 cm, height 6 cm: area = ½ × 6 × (9 + 15) = **72 cm²**.
- Derivation 1: cut it into a rectangle and two triangles and add the three areas.
- Derivation 2: put two copies together to make a parallelogram of base (a + b) and height h; the trapezium is half of it.

Shape	Area formula	Example
Rectangle	l × w	9 × 6 = 54
Triangle	½ × b × h	½ × 12 × 7 = 42
Parallelogram	b × h	15 × 8 = 120
Rhombus	½ × d_1 × d_2	½ × 16 × 12 = 96
Trapezium	½ × h × (a + b)	½ × 6 × 24 = 72

Units of Area in Real Life

Conversion	Value
1 in	2.54 cm
1 in²	2.54^2 = 6.4516 cm²
10 in²	64.516 cm²
161.29 cm² in in²	161.29 ÷ 6.4516 = 25 in²
1 ft	12 in, so 1 ft² = 144 in²
1 m²	10,000 cm²
1 km²	10,00,000 m²

Conversion	Value
1 acre	43,560 ft²

- An A4 sheet is 21 cm × 29.7 cm = **623.7 cm²**.
- Local units: bigha, gaj, katha, dhur, cent, ankanam. Large regions are measured in **km²**.

> **NOTE**
> When converting areas, square the length factor: 1 ft = 12 in gives 1 ft² = 12 × 12 = 144 in², not 12 in².

Common Mistakes

Mistake	Correct
Using the slant side as the height of a parallelogram	Use the perpendicular height
Forgetting ½ in the triangle, rhombus or trapezium formula	Triangle and rhombus have ½ × …; trapezium has ½ × h × (a + b)
Writing area in cm	Write cm² (square units)
Converting 1 ft² as 12 in²	1 ft² = 144 in²

Likely question	Model answer
Can perimeter measure area?	No. Different shapes with the same perimeter have different areas, and a larger perimeter can have a smaller area.
What is the area of a triangle?	½ × base × height, with the height perpendicular to the base.
How do we find the area of any polygon?	Divide it into triangles and add their areas.
What is the area of a rhombus with diagonals 16 cm and 12 cm?	½ × 16 × 12 = 96 cm².
Why do all triangles on a common base between two parallel lines have equal area?	They have the same base and the same height, so ½ × base × height is the same.

★ KEY TAKEAWAYS

- Area of a rectangle = length × width; perimeter is not a measure of area.
- Triangle: ½ × base × height; a median splits a triangle into two equal-area triangles.
- Triangles on the same base between parallel lines have equal area.
- Any polygon can be divided into triangles; quadrilateral = ½ × diagonal × (sum of perpendiculars).
- Parallelogram = base × height; rhombus = ½ × d_1 × d_2; trapezium = ½ × h × (a + b).
- 1 in² = 6.4516 cm²; 1 ft² = 144 in²; 1 km² = 10,00,000 m²; 1 acre = 43,560 ft².

CHAPTER 10 · MATHS (PART 2)

Exploring Some Geometric Themes

This chapter has two themes. First, fractals: shapes that repeat the same pattern at smaller and smaller scales, which we study by counting how squares, triangles, holes and sides grow at each step. Second, visualising solids: faces, edges and vertices, nets, shortest paths on a cuboid, and drawing solids using front, top and side views and isometric grids.

Big Questions

- What is a fractal and how do we count its parts step by step?
- How do nets help us make solids and find shortest paths?
- How do we represent a solid on a flat page?

Fractals and Self-Similarity

DEFINITION

A **fractal** is a self-similar shape: it shows the same or a similar pattern over and over again at smaller and smaller scales. It is built by repeating one rule at each **step**.

- In nature: ferns, trees, clouds, coastlines, mountains, lightning.
- In art: temple towers (for example Kandariya Mahadev Temple, Khajuraho, c. 1025 CE) made of smaller copies of the whole; Fulani wedding blankets; M. C. Escher's 'Smaller and Smaller'.

Sierpinski Carpet

Start with a square (Step 0). Divide it into **9** equal squares and remove the **central** one. Repeat the same on each remaining square.

Step n	Squares remaining R_n	Holes H_n	Area left (start = 1)
0	1	0	1
1	8	1	8/9
2	64	1 + 8 = 9	64/81
3	512	1 + 8 + 64 = 73	512/729

FORMULA

$R_{n+1} = 8R_n$, so $\mathbf{R_n = 8^n}$. $H_{n+1} = H_n + R_n$, so $H_n = 1 + 8 + 8^2 + \ldots + 8^{n-1} = \mathbf{(8^n - 1) \div 7}$. Area left = $\mathbf{(8/9)^n}$.

Sierpinski Gasket (Triangle)

Start with an equilateral triangle. Join the midpoints of the sides to get **4** identical equilateral triangles and remove the **central** one. Repeat on each remaining triangle.

Step n	Triangles remaining	Holes	Area left (start = 1)
0	1	0	1

Step n	Triangles remaining	Holes	Area left (start = 1)
1	3	1	3/4
2	9	1 + 3 = 4	9/16
3	27	1 + 3 + 9 = 13	27/64

FORMULA

Triangles = $\mathbf{3^n}$; holes = 1 + 3 + … + 3^{n-1} = $\mathbf{(3^n - 1) \div 2}$; area left = $\mathbf{(3/4)^n}$.

NOTE

The four small triangles are identical because the midpoint segments are half the sides, so all four triangles have three equal sides.

Koch Snowflake

Start with an equilateral triangle. Divide each side into **3** equal parts, raise an equilateral triangle on the middle part and remove that middle part. Repeat on every side of the new shape.

Step n	Number of sides	Length of each side (start side 1)	Perimeter
0	3	1	3
1	12	1/3	4
2	48	1/9	16/3
3	192	1/27	64/9

FORMULA

Sides = $\mathbf{3 \times 4^n}$; each side = $(1/3)^n$; perimeter = $\mathbf{3 \times (4/3)^n}$. The perimeter keeps growing, because each side is replaced by 4 parts that are each one-third as long.

Solids: Faces, Edges, Vertices

- **Faces** are the flat surfaces, **edges** are the line segments where faces meet, **vertices** are the corner points.
- A cuboid or cube has **6 faces, 12 edges, 8 vertices**.
- A **prism** has two congruent polygon faces joined by parallelograms; a **pyramid** has a polygon base and a point (apex) joined to every vertex; a triangular pyramid is a **tetrahedron**.

Solid with n-sided base	Faces	Edges	Vertices	n = 10
Prism	n + 2	3n	2n	12, 30, 20
Pyramid	n + 1	2n	n + 1	11, 20, 11

IMPORTANT

For these solids **faces + vertices − edges = 2**: prism with n = 10 gives 12 + 20 − 30 = 2; pyramid gives 11 + 11 − 20 = 2.

Nets of Solids

DEFINITION

A **net** is the flat shape obtained by unfolding a solid onto a plane; it can be folded back into the solid (extra flaps for sticking are not part of the net).

Solid	Net / facts
Cube	6 squares; **11** different nets (rotations and flips counted as one)
Cuboid	6 rectangles in 3 equal pairs, e.g. 5 × 3, 5 × 1, 3 × 1 for a 5 × 3 × 1 cuboid
Regular tetrahedron	4 equilateral triangles; **2** nets
Octahedron	8 equilateral triangles; **11** nets
Dodecahedron	12 pentagons; **43,380** nets
Cylinder	2 circles and a rectangle of width $2\pi r$ and height h
Cone	1 circle and a sector of a larger circle
Sphere	**No net** — a flat paper cannot wrap a ball without wrinkles or gaps

Example: for a cylinder of radius 7 cm and height 10 cm ($\pi = 22/7$) the rectangle in the net is $2 \times 22/7 \times 7 =$ **44 cm** by **10 cm**.

Shortest Paths on a Cuboid

On a plane the shortest path is a straight line. On a cuboid surface, unfold it into a net: a path on the solid has the **same length** as the matching path on the net, so the shortest path is the **straight line on a suitable net**.

Step 1
List the different ways of unfolding the cuboid so that both points lie on the net.

Step 2
Join the two points by a straight line on each net; the line must stay inside the net.

Step 3
Find each length using the Baudhāyana (Pythagoras) theorem.

Step 4
The smallest length is the shortest path.

- If an unfolding gives legs 24 and 32, the path is $\sqrt{(24^2 + 32^2)} =$ **40**; legs 9 and 12 give **15**.
- Different unfoldings give different lengths, so **the way the cuboid is unfolded matters**; always compare them all.

Projections and Views

- A solid seen from different viewpoints has different outlines: a cylinder looks like a rectangle from the side and a circle from above; a cone like a triangle from the side and a circle from below.
- We describe a solid with its **front view** (vertical plane), **top view** (horizontal plane) and **side view** (side plane).
- A shadow under a very distant light, such as the Sun shining perpendicular to a plane, is the same as the **projection** on that plane.

Isometric Drawings

- Orient a cube so that all its edges appear equally long; this is an **isometric projection**.
- Draw it on **isometric grid paper** (triangle dots at 60°); edges along the three directions of the grid keep their true lengths.
- Different solids made of identical cubes can be drawn and compared on the isometric grid.

Common Mistakes

Mistake	Correct
Writing $H_n = 8^n$ for the carpet	$R_n = 8^n$ are the squares; holes $H_n = (8^n - 1) \div 7$
Forgetting that a Koch snowflake's perimeter grows	Perimeter = $3 \times (4/3)^n$ increases at each step
Using one unfolding for the shortest path	Try every unfolding and take the smallest
Counting an ant's path that leaves the net	A line leaving the net is not a path on the cuboid

Likely question	Model answer
What is a fractal?	A self-similar shape that repeats the same pattern at smaller and smaller scales.
How many squares remain at Step n of the Sierpinski carpet?	8^n.
How many sides does the Koch snowflake have at Step n?	3×4^n.
Why does a sphere have no net?	A flat sheet cannot cover a sphere without wrinkles, gaps or overlaps.
How is the shortest path on a cuboid found?	Unfold the cuboid into a suitable net and join the two points by a straight line that stays inside the net.

★ KEY TAKEAWAYS

- Fractals repeat one rule at smaller scales; they occur in nature and in art.
- Carpet: squares 8^n, holes $(8^n - 1) \div 7$, area $(8/9)^n$.
- Gasket: triangles 3^n, holes $(3^n - 1) \div 2$, area $(3/4)^n$.
- Koch snowflake: sides 3×4^n, perimeter $3 \times (4/3)^n$.
- A cube has 11 nets, a regular tetrahedron 2, an octahedron 11; a sphere has none.
- Prism with n-gon base: n + 2 faces, 3n edges, 2n vertices; pyramid: n + 1, 2n, n + 1.
- Shortest path on a cuboid = straight line on the best net; check every unfolding.
- Describe solids by front, top and side views and by isometric drawings.

CHAPTER 11 · MATHS (PART 2)

Fractions in Disguise

A percentage is a fraction with denominator 100. This chapter converts between fractions, decimals and percentages, finds a percentage of a quantity, handles percentages above 100, percentage change, profit, loss, discount and tax, and compares growth with and without compounding.

Big Questions

- How are fractions, decimals and percentages related?
- How do we use percentages for change, profit, loss, discount and tax?
- What is the difference between interest with and without compounding?

Fractions as Percentages

DEFINITION

Per cent means ‘out of hundred’ (Latin *per centum*). **x% = x/100**, so 25% means 25 out of every 100.

- To change a fraction into a percentage, **multiply by 100**: 3/5 = 3/5 × 100 = **60%**; 7/20 = **35%**.
- To change a percentage into a fraction, write it over 100 and simplify: 24% = 24/100 = **6/25**.
- Percentages are just fractions with a **common denominator 100**, which makes comparison easy (a quick glance shows 28% is more than 26%).

Percent	Fraction	Decimal
50%	1/2	0.5
25%	1/4	0.25
75%	3/4	0.75
10%	1/10	0.1
5%	1/20	0.05
1%	1/100	0.01
43%	43/100	0.43
100%	1	1.0

NOTE

Per 100 is used because our number system is base 10, so decimals and percentages convert quickly. The idea of ‘per hundred’ appears in Kautilya’s *Arthaśāstra* (4th century BCE) in the rates of interest.

Percentage of a Quantity

FORMULA

x% of a quantity y = (x/100) × y. Example: 35% of 240 = 0.35 × 240 = **84**.

- 25% of a number = ¼ of it; 50% = ½; 10% = one-tenth.
- **Mental tricks:** 20% = 2 × 10%; 5% = half of 10%; 25% = 20% + 5%; 15% = 10% + 5%; 75% = 100% − 25%.
- For 240: 10% = 24, 5% = 12, 15% = 36, 25% = 60.

Comparing percentages of **different wholes** needs care. 25% of 120 g of biscuits = 30 g sugar; 35% of 95 g = 33.25 g. So the 35% biscuits have more sugar here only because we computed the real amounts.

▸ Finding the whole from a part

A cyclist has covered 30% of a journey, which is 66 km. How much remains?

Step 1 — Diagram
Draw a bar of 100%; mark 30% = 66 km.

Step 2 — Find 1% or the whole
30% = 66 km, so 10% = 22 km and 100% = **220 km**.

Step 3 — Remaining
220 − 66 = **154 km** (or 70% of 220 = 154).

IMPORTANT

Drawing a rough bar model first helps you see the situation and estimate before calculating.

Ratios and Percentages

Millet and water mixed in the ratio **3 : 7** make 10 parts. Millet = 3/10 = **30%** and water = **70%**. In 500 ml of mixture, millet = 30% of 500 = **150 ml**.

Percentages Greater than 100

- A percentage above 100 means the quantity is **more than the original**: 125% = 1.25 times.
- Target of ₹8,000 and sales of ₹10,000: 10000/8000 × 100 = **125%** of the target, i.e. 25% more than the target.
- 200% means **double**; 250% means 2.5 times: 250% of ₹8,000 = ₹20,000.
- Wheat 260 kg last year, 650 kg this year: 650/260 × 100 = **250%** of last year.

Percentage Increase or Decrease

FORMULA

Percentage change = (amount of change ÷ original amount) × 100. The base is always the **original** amount.

Situation	Change	Percentage
Price ₹40 → ₹46	+ ₹6	6/40 × 100 = **15% increase**
Footfall 250 → 200	− 50	50/250 × 100 = **20% decrease**

- ‘New value is 165% of the old’ and ‘increased by 65%’ say the same thing: new = old × 1.65.
- Increase by x%: multiply by **(1 + x/100)**; decrease by x%: multiply by **(1 − x/100)**.
- ‘Demand fell by 85%’ means demand now is **15%** of the earlier demand.

Profit and Loss

Term	Meaning
Cost price (CP)	Price at which the seller bought the item
Marked price (MP)	Price quoted or printed on the item
Selling price (SP)	Price actually paid by the customer
Profit	SP − CP when SP > CP
Loss	CP − SP when SP < CP
Discount	Reduction from the marked price

FORMULA

Profit % = (profit ÷ CP) × 100; Loss % = (loss ÷ CP) × 100. Percentages are taken on the **cost price** unless the question says otherwise.

Example	Working	Answer
CP ₹400, SP ₹500	profit 100; 100/400 × 100	Profit **25%**
CP ₹800, SP ₹680	loss 120; 120/800 × 100	Loss **15%**
CP ₹2000, loss 12%	SP = 88% of 2000	SP **₹1760**
SP ₹1100, profit 10%	SP = 110% of CP, so CP = 1100 ÷ 1.1	CP **₹1000**
MP ₹1500, 20% discount, CP ₹1000	SP = 80% of 1500 = 1200; profit 200/1000	Profit **20%**

- Profit as a percentage of **sales** is different: CP ₹400 and SP ₹500 give 25% on cost but 100/500 × 100 = **20% of sales**.
- Gross profit = sales − cost of goods; net profit = gross profit − other expenses (transport, salary, electricity).
- Two articles sold at ₹2100 (5% profit, CP ₹2000) and ₹1800 (10% loss, CP ₹2000): total CP ₹4000, total SP ₹3900, so **overall loss ₹100 = 2.5%**.

Taxes

Tax rates such as **GST** are percentages added to the price. On goods priced ₹2,500 with GST at 12%: tax = 12% of 2500 = ₹300, so bill = 2500 + 300 = **₹2,800**.

Growth and Compounding

DEFINITION

Interest is extra money paid on a deposit or loan. **Principal (p)** is the money deposited; **r** is the rate per year (as a decimal); **p.a.** means per annum (each year).

- Fixed Deposit: money stays with the bank for a fixed term; the interest is paid either regularly (no compounding) or added back (compounding).

	Without compounding	With compounding
Idea	Interest is paid out; principal stays the same	Interest is added back; the principal grows every term
Amount after t terms	**$p(1 + rt)$**	**$p(1 + r)^t$**
Growth	Linear (same interest each year)	Exponential (interest on interest)
₹8000 at 10% p.a., 3 years	8000 × 1.3 = **₹10,400**	8000 × 1.1 × 1.1 × 1.1 = 8000 × 1.331 = **₹10,648**

Step 1 — Year 1
8000 × 1.1 = ₹8,800.

Step 2 — Year 2
8800 × 1.1 = ₹9,680.

Step 3 — Year 3
9680 × 1.1 = ₹10,648, a gain of 33.1% against 30% without compounding.

IMPORTANT

Compounding always gives **more** than simple growth after the first term, and the gap widens with time.

Decline (Depreciation)

Items lose value with age and use, called **depreciation**. A fall of x% per period multiplies the value by **$(1 - x/100)$** each period.

- Scooter worth ₹60,000 depreciating 10% a year: after 2 years = 60000 × 0.9 × 0.9 = **₹48,600**.
- Village population 2000 falling 5% per decade: after 2 decades = 2000×0.95^2 = **1805**.
- Round only at the end, since rounding at each step changes the result a little.

Common Mistakes

Mistake	Correct
Taking the percentage change on the new value	Use the original amount as the base
Adding 10% twice to get a 20% compound rise	Two years at 10% compounded = 1.1 × 1.1 = 1.21, i.e. 21%
Calculating discount on the cost price	Discount is on the marked price
Reading '165% of' as '65% more' differently	165% of old = old + 65% of old

Likely question	Model answer
What does 35% mean?	35 out of every 100, i.e. 35/100 = 0.35.
How do we change 7/20 into a percentage?	7/20 × 100 = 35%.
What is the formula for percentage change?	(change ÷ original) × 100.
Write the amounts with and without compounding.	Without: $p(1 + rt)$; with: $p(1 + r)^t$.

Likely question	Model answer
A fall of 20% followed by a rise of 25%: is the value unchanged?	Yes here: 0.8 × 1.25 = 1, so it returns to the original. A rise of 20% followed by a fall of 20% does not: 1.2 × 0.8 = 0.96.

★ KEY TAKEAWAYS

- x% = x/100; a fraction becomes a percentage on multiplying by 100.
- x% of y = (x/100) × y; use 10%, 5%, 25% tricks for mental sums.
- Percentages can exceed 100; 200% means double.
- Percentage change = (change ÷ original) × 100; increase multiplies by (1 + x/100), decrease by (1 − x/100).
- Profit % or loss % is on the cost price; discount is on the marked price.
- Without compounding: $p(1 + rt)$; with compounding: $p(1 + r)^t$, which is always larger for $t > 1$.
- Depreciation and population decline repeat the factor (1 − x/100) each period.

CHAPTER 12 · MATHS (PART 2)

Proportional Reasoning – 2

When quantities change by the same factor they are proportional. This chapter extends ratios to three or more terms, uses them to read maps, divide a whole and draw pie charts, and then meets inverse proportion, where one quantity grows exactly as fast as the other shrinks.

Big Questions

- How do we work with ratios having more than two terms?
- How can a whole be divided in a given ratio and shown as a pie chart?
- How are direct and inverse proportion different, and how do we solve problems on each?

Recap: Proportional Ratios

DEFINITION

Two ratios **a : b** and **c : d** are **proportional** (written a : b :: c : d) if **a × d = b × c**, that is, a/c = b/d.

- Example: 9 : 6 and 6 : 4 are proportional because 9 × 4 = 36 and 6 × 6 = 36.
- Proportional ratios are *equivalent ratios*: both terms change by the same factor (here 9 : 6 becomes 6 : 4 when both are multiplied by 2/3).
- A ratio is in its **simplest form** when its terms have no common factor greater than 1 (divide every term by their HCF).

Ratios in Maps

DEFINITION

The **Representative Fraction (RF)** of a map is the ratio **map distance : actual (geographical) distance**. RF 1 : 50,00,000 means 1 cm on the map stands for 50,00,000 cm on the ground.

Step 1 — Measure
Measure the distance between the two places on the map with a ruler (in cm).

Step 2 — Multiply
Multiply by the second term of the RF to get the actual distance in cm.

Step 3 — Convert
Change cm to km: 1 km = 1,00,000 cm, so divide by 1,00,000.

RF 1 : 50,00,000 means 1 cm = 50,00,000 cm = **50 km**. A map distance of 4.6 cm is therefore 4.6 × 50 = **230 km**. This is the **straight-line (geographical)** distance, not the road distance.

NOTE

Maps of different scales give the same real distance. A map with a bigger second term (e.g., 1 : 1,00,00,000) shows the same place with a shorter map distance.

Ratios with More than Two Terms

A ratio such as **8 : 4 : 2 : 1** tells us that for every 8 units of the first quantity there are 4, 2 and 1 units of the others. If every quantity is multiplied (or divided) by the same factor, the new ratio is **proportional** to the old one.

> **FORMULA**
> If **a : b : c : d :: p : q : r : s**, then **a/p = b/q = c/r = d/s**.

▸ Worked example — mixed nuts

A snack mix has almonds : cashews : raisins :: 3 : 2 : 5. If we use 40 g of cashews, one part = 40 ÷ 2 = 20 g. Almonds = 3 × 20 = **60 g**, raisins = 5 × 20 = **100 g**, and the whole mix = 60 + 40 + 100 = **200 g**.

Dividing a Whole in a Given Ratio

> **FORMULA**
> To divide a quantity **x** in the ratio **a : b : c**, the parts are **x × a/(a+b+c)**, **x × b/(a+b+c)** and **x × c/(a+b+c)**. The parts always add up to x.

Step 1
Add the terms of the ratio to find the total number of parts.

Step 2
Divide the whole by the total parts to get the value of one part.

Step 3
Multiply each term by the value of one part.

Step 4 — Check
Add the parts; the sum must equal the whole.

Problem	Total parts	One part	Answer
Divide 240 in the ratio 2 : 3 : 5	10	24	48, 72, 120
Angles of a triangle in the ratio 2 : 3 : 4	9	180° ÷ 9 = 20°	40°, 60°, 80°
₹1,500 shared in the ratio 1 : 2 : 3	6	₹250	₹250, ₹500, ₹750

> **IMPORTANT**
> Not every ratio can be the sides of a triangle. The two shorter sides must add up to **more than** the longest side: 3 : 4 : 5 works (3 + 4 > 5) but 1 : 3 : 5 does not (1 + 3 < 5).

A Slice of the Pie: Pie Charts

A **pie chart** shows how a whole is shared among categories. Each slice is a **sector** whose angle is proportional to its share, and the angles add up to **360°**.

Step 1
Write the data as a ratio and reduce it to its simplest form (divide by the HCF).

Step 2
Divide 360° in this ratio to get the angle of each slice.

Step 3
Draw a circle and one radius; with a protractor measure the first angle from this radius.

Step 4
Start each new slice from where the last one ended; colour and label the slices.

Example. 30 students chose a sport: cricket 10, football 8, kabaddi 6, hockey 4, chess 2. The ratio 10 : 8 : 6 : 4 : 2 reduces to **5 : 4 : 3 : 2 : 1** (total 15 parts). One part = 360° ÷ 15 = 24°.

Sport	Students	Angle
Cricket	10	5 × 24° = 120°
Football	8	4 × 24° = 96°
Kabaddi	6	3 × 24° = 72°
Hockey	4	2 × 24° = 48°
Chess	2	1 × 24° = 24°
Total	30	360°

NOTE

If the data is given in **percentages**, the angle of a slice = percentage × 360° ÷ 100 = percentage × 3.6°. A pie chart can also be read backwards: if 18 children form a slice of 60°, the whole group is 18 × 6 = 108 children, because 360° = 6 × 60°.

Direct Proportion

Two quantities are **directly proportional** when they change by the **same factor in the same direction** and their quotient stays constant: $\mathbf{x_1/y_1 = x_2/y_2 = k}$. This is the *rule of three*: if a : b :: c : d, then **d = (b × c) ÷ a**.

- 24 pencils cost ₹120, so 20 pencils cost 120 × 20 ÷ 24 = **₹100**.
- More workers moving bricks in a day means more bricks moved: direct proportion.
- Cloth length and price at a fixed rate per metre are directly proportional.

Inverse Proportion

DEFINITION

Two quantities **x** and **y** are in **inverse proportion** if **x × y = k** (a constant). When one changes by a factor n, the other changes by the factor **1/n**.

Speed (km/h)	5	15	30	60
Time for 90 km (hours)	18	6	3	1.5
Product (speed × time)	90	90	90	90

FORMULA

If x and y are inversely proportional: $\mathbf{x_1 \times y_1 = x_2 \times y_2}$, or $\mathbf{x_1/x_2 = y_2/y_1}$.

▸ Worked example — workers and days

12 workers finish a job in 10 days. How long will 15 workers take? The quantities are inversely proportional, so 12 × 10 = 15 × y, giving y = 120 ÷ 15 = **8 days**.

▸ Worked example — provisions

Food for 80 students lasts 15 days. If 20 more join, the stock lasts 80 × 15 ÷ 100 = **12 days**.

Direct proportion	Inverse proportion
Both increase or both decrease	One increases, the other decreases
x/y is constant	x × y is constant
Cost and number of articles; distance and time at fixed speed	Speed and time for a fixed distance; workers and days for a fixed job

> **IMPORTANT**
> Always state the **assumptions**: all workers or pumps work at the same rate, the job is fixed, and the food is eaten at the same rate. Without them the proportion may not hold.

Working Together

Treat the whole job as **1 unit of work**. If A finishes it in 2 hours and B in 3 hours, then in one hour A does 1/2 and B does 1/3, so together they do 1/2 + 1/3 = **5/6** of the work per hour. Time for the whole job = 1 ÷ 5/6 = **6/5 hours = 1 h 12 min**.

> **REMEMBER**
> Add the **rates** (work per hour), not the times. Two pumps that fill a tank in 3 h and 2 h do **not** take 5 hours together; together they take less than the faster pump alone.

Common Mistakes

- Using direct proportion when the quantities move in opposite directions (more workers, fewer days).
- Forgetting to convert units (cm to km) in map problems.
- Leaving the ratio unsimplified before finding pie-chart angles; the angles still work, but the arithmetic is harder.
- Dividing a whole by one term of the ratio instead of by the **sum** of the terms.

Likely question	Model answer
What does RF 1 : 20,00,000 mean?	1 cm on the map represents 20,00,000 cm = 20 km on the ground.
How do you divide 180 in the ratio 1 : 2 : 3?	Total parts = 6, one part = 30, so the shares are 30, 60 and 90.
How is the angle of a pie-chart slice found?	Angle = (value ÷ total) × 360°; the angles of all slices add up to 360°.
If 6 taps fill a tank in 8 hours, how long will 4 taps take?	Inverse proportion: 6 × 8 = 4 × y, so y = 12 hours.
How can you test whether two quantities are in inverse proportion?	Multiply corresponding values; if the product is the same every time, they are inversely proportional.

★ KEY TAKEAWAYS

- Ratios a : b : c : … show how many units of each quantity go together; proportional ratios have equal term-by-term factors.
- RF = map distance : actual distance; convert to km by dividing cm by 1,00,000.
- To divide x in the ratio a : b : c, give each part x × (its term) ÷ (sum of the terms).
- Pie-chart slice angle = (share of the whole) × 360°.
- Direct proportion: x/y is constant. Inverse proportion: x × y is constant.
- Work-together problems: add the work done per hour, then take the reciprocal.
- Check the assumptions and units before using a proportion.

CHAPTER 13 · MATHS (PART 2)

Tales by Dots and Lines

The mean is the balance point of data, and the median is its middle. This chapter shows how both react when values are added, removed or changed, how to find missing values, how to use frequency tables and spreadsheet formulae, and how to read line graphs and infographics sensibly.

Big Questions

- How does the mean behave as a balance point, and what happens when the data changes?
- How do we find a mean or median from a frequency table or a spreadsheet?
- How do we read and interpret line graphs and other data displays?

The Mean as a Balancing Point

DEFINITION

Mean (average) = sum of all values ÷ number of values. The **median** is the middle value of the data arranged in order (the average of the two middle values if the number of values is even).

- On a dot plot, the mean is the point where the data **balances**: the **total distance** of the values below the mean equals the **total distance** of the values above it.
- The mean of two numbers is exactly halfway between them (3 and 7 give 5).
- The mean is **not** always the midpoint of the smallest and largest values. For 10, 10, 11, 17 the mean is 12, but the midpoint of 10 and 17 is 13.5.
- There is only **one** such balance point: shifting it in either direction makes one side's distance larger than the other's.

What Happens to the Mean When Data Changes?

Change	Effect on the mean	Reason
Include a value greater than the mean	Mean increases	More distance on the upper side, so the balance point moves up
Include a value smaller than the mean	Mean decreases	Balance point moves down
Include or remove a value equal to the mean	Mean stays the same	It adds no distance on either side
Remove a value greater than the mean	Mean decreases	Upper-side distance shrinks
Add a fixed number k to every value	Mean increases by k	New mean = a + k
Multiply every value by k	Mean is multiplied by k	New mean = k × a

FORMULA

If $x_1, x_2, \ldots x_n$ have mean **a**, then the values $x_1 + k, \ldots x_n + k$ have mean **a + k**, and the values $kx_1, \ldots kx_n$ have mean **k × a**.

Proof for adding k. New mean = $[(x_1 + k) + (x_2 + k) + \ldots + (x_n + k)] \div n = (x_1 + x_2 + \ldots + x_n + nk) \div n = a + k$. **Check:** the data 4, 6, 7, 7, 9 has mean 6.6. Adding 10 to every value gives mean 16.6; doubling every value gives mean 13.2.

NOTE

Several values can be added together without changing the mean if their distances from the mean cancel out: for example, two values below the mean and one above it, chosen so that the total distance below equals the distance above.

What Happens to the Median?

- Including a value **greater** than the median makes the median **increase** (or stay the same); including a smaller value makes it decrease (or stay the same).
- Example: for 5, 6, 8, 8, 9, 10 the median is 8. Including 11 gives seven values and the median becomes the 4th value, 8; including two large values moves the middle to the right.
- The median depends on the **position** of values, so extreme values change it much less than they change the mean.

Finding an Unknown Value

Step 1
Use total = mean × number of values.

Step 2
Add the known values.

Step 3
Unknown value = total − sum of the known values.

Example. The mean of 8, 10, 12, 15, 9 and one smudged number is 12. Total of six values = 12 × 6 = 72. The known five add up to 54, so the missing value is 72 − 54 = **18**.

Correcting a mean. A shopkeeper finds the mean daily sale over 20 days is ₹18.5 (in thousands), so the total is 370. One day was entered 5 too high, so the correct total is 365 and the correct mean is 365 ÷ 20 = **18.25**. We did not need the individual values.

Mean and Median from a Frequency Table

When a value occurs several times, multiply it by its **frequency** before adding. Do not average the distinct values alone.

Family size	3	4	5	6
Frequency (students)	2	5	8	5
Value × frequency	6	20	40	30

- Number of values = 2 + 5 + 8 + 5 = **20**; sum = 6 + 20 + 40 + 30 = **96**.
- Mean = 96 ÷ 20 = **4.8**.

- Median: the 10th and 11th values are the middle. Cumulative frequencies are 2, 7, 15, so both lie in the group of 5. Median = **5**.

> **IMPORTANT**
> To find the median from a frequency table, add the frequencies **cumulatively** from the smallest value until you reach the middle position. You never have to write out all the values.

Spreadsheets

A **spreadsheet** is a grid of **cells** arranged in **columns** (A, B, C, …) and **rows** (1, 2, 3, …). A cell is named by its column letter and row number, so E5 is column E, row 5.

Formula	What it does
=SUM(B3:G3)	Adds the marks in cells B3 to G3 (one student's total across subjects)
=AVERAGE(D2:D6)	Mean of the first five students' marks in column D
=AVERAGE(G2:G23)	Class average of a column of 22 students

- **Start:End** (for example B3:G3) names a block of cells in one row or column.
- A formula always begins with **=**.
- Change one number and every formula that uses it updates automatically.

Line Graphs

> **DEFINITION**
> A **line graph** joins data points with line segments. It is best for showing how a quantity **changes over time**, for example the monthly maximum temperature of a state.

Step 1 — Identify what is given
Read the title, the axes, the scale, the units and the legend (colours or marker shapes for each line).

Step 2 — Infer and interpret
Describe the trend (rising, falling, flat), find peaks and low points, compare lines, and note the steepness: a steeper segment means a faster change.

Step 3 — Question the data
Ask how the data was collected, what is missing, and what can and cannot be concluded.

- A clustered column graph with 13 years × 4 countries would need 52 bars and look crowded; a line graph shows the same trend clearly.
- Monthly *averages* (for example rainfall) are obtained by averaging the same month over several years.
- Different shaped markers help readers who print in black and white or find colours hard to distinguish.
- Use a line graph for trends over time and a bar graph for comparing separate categories.

▸ Example reading

If a line for City A rises from 20 to 38 between January and June and then falls, while City B stays between 29 and 33 all year, we can say City A's temperature varies much more than City B's. The graph tells us what happens, not why; the reasons need further questions and data.

Other Ways of Showing Data

- **Infographic:** combines colours, maps and symbols to compare data at a glance, for example a map shaded by how strongly each state prefers rice or wheat. Read the colour scale first.
- **Time strip:** a strip of 48 boxes (one for each half hour from midnight to midnight) coloured by activity; it shows how a day is spent.
- **Data story:** a graph plus questions, such as how daily sleep changes with age; a smooth curve is used because it has many closely spaced points.

Common Mistakes

- Averaging only the distinct values of a frequency table instead of multiplying by frequencies.
- Taking the mean to be the midpoint of the smallest and largest values.
- Forgetting that the median needs the data to be **sorted** first.
- Reading a line graph without checking the scale or where the axis starts.

Likely question	Model answer
Why is the mean called a balance point?	The total distance of the values below it equals the total distance of the values above it.
What happens to the mean of a set if every value increases by 5?	The mean also increases by 5.
What happens to the mean if a value equal to the mean is added?	It does not change.
The mean of 5 numbers is 14. Four of them add up to 58. What is the fifth?	Total = 70, so the fifth number is 70 − 58 = 12.
What does =AVERAGE(C2:C10) compute?	The mean of the nine numbers in cells C2 to C10.

★ KEY TAKEAWAYS

- Mean = sum ÷ number of values; it is the balance point where distances on both sides are equal.
- Adding a value above the mean raises it, below the mean lowers it, equal to the mean leaves it unchanged.
- Adding k to every value adds k to the mean; multiplying every value by k multiplies the mean by k.
- Missing values: total = mean × number of values, then subtract the known values.
- For a frequency table, mean = sum of (value × frequency) ÷ total frequency; find the median by cumulative frequency.
- Spreadsheet cells are named by column letter and row number; =SUM(…) and =AVERAGE(…) work on ranges like B3:G3.
- Read line graphs in two steps: first what is given (axes, scale, legend), then what it means; line graphs suit trends over time.

CHAPTER 14 · MATHS (PART 2)

The Baudhāyana–Pythagoras Theorem

In a right-angled triangle the square on the longest side equals the sum of the squares on the other two sides. This chapter builds the idea from doubling and halving squares, finds the length of the diagonal of a square, studies whole-number triples such as (3, 4, 5) and applies the theorem to solve problems.

Big Questions

- How can we make a square with double or half the area of a given square?
- What relation connects the three sides of a right-angled triangle?
- How can the theorem be used to find unknown lengths and to generate whole-number triples?

Doubling and Halving a Square

- Doubling the side of a square makes the area **4 times**, not 2 times.
- **Baudhāyana's rule (Śulba-Sūtra, c. 800 BCE):** the square built on the **diagonal** of a square has **double** the area of the original square.
- Reason: the original square is made of 2 congruent right triangles, and the square on its diagonal is made of 4 of the same triangles.
- To **halve** a square, join the midpoints of its sides; the tilted square inside has **half** the area. Halving the side gives a square of only one-quarter the area.

Isosceles Right Triangle and √2

FORMULA

In an isosceles right triangle with equal sides **a** and hypotenuse **c**: $\mathbf{c^2 = 2a^2}$, so $\mathbf{c = a\sqrt{2}}$.

Given	Working	Result
a = 12	c = √(2 × 144) = √288; $16^2 = 256$ and $17^2 = 289$	c is between 16 and 17
a = 5	c = √50; $7.07^2 = 49.98$ and $7.08^2 = 50.13$	7.07 < c < 7.08
c = √98	$2a^2 = 98$, $a^2 = 49$	a = 7

DEFINITION

The **hypotenuse** is the side opposite the right angle, and it is the **longest side** of a right-angled triangle.

▸ Facts about √2

- 1 < √2 < 2, and closer bounds: 1.4 < √2 < 1.5, 1.41 < √2 < 1.42, **1.414 < √2 < 1.415**; √2 = 1.41421356…
- √2 cannot be a **terminating decimal**: a terminating decimal ending in a non-zero digit has a square ending in a non-zero digit, so its square can never be exactly 2.000…
- √2 cannot be written as a fraction m/n of counting numbers. If $2n^2 = m^2$, the prime 2 would appear an odd number of times on the left and an even number of times on the right. (Proof by Euclid, c. 300 BCE.)
- So √2 has a **non-terminating** decimal expansion.

Combining Two Different Squares

Baudhāyana (Verse 1.12): *the square on the diagonal produces the sum of the squares on the two sides.* To combine two squares of sides a and b, make a right-angled triangle whose perpendicular sides are a and b. The square on its hypotenuse has area equal to $a^2 + b^2$.

FORMULA

Baudhāyana–Pythagoras Theorem: in a right-angled triangle with perpendicular sides **a** and **b** and hypotenuse **c**, $\mathbf{a^2 + b^2 = c^2}$.

Proof idea: four congruent right triangles with sides a, b, c are placed around a square so that the four-sided figure formed is a square of side c. Its area equals the sum of the areas of the two original squares.

Step 1
Identify the hypotenuse (opposite the right angle) and name it c.

Step 2
Write $a^2 + b^2 = c^2$ with the correct side as c.

Step 3
Substitute the known values and solve for the unknown square.

Step 4
Take the square root; leave an irrational answer between two whole numbers if needed.

a	b	c	Working
3	4	5	9 + 16 = 25
5	12	13	25 + 144 = 169
7	?	25	$b^2 = 625 - 49 = 576$, b = 24
8	15	17	64 + 225 = 289

IMPORTANT

The theorem applies **only to right-angled triangles**. The hypotenuse is always the **longest** side, so it cannot be one of the shorter sides in $a^2 + b^2 = c^2$.

Baudhāyana Triples

DEFINITION

A **Baudhāyana (Pythagorean) triple** (a, b, c) is a set of positive integers with $\mathbf{a^2 + b^2 = c^2}$. A triple with no common factor greater than 1 is **primitive**.

- Baudhāyana listed (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), (12, 35, 37) and (15, 36, 39).
- All triples with numbers up to 20: (3, 4, 5), (6, 8, 10), (9, 12, 15), (12, 16, 20), (5, 12, 13), (8, 15, 17). The primitive ones are (3, 4, 5), (5, 12, 13) and (8, 15, 17).
- **Scaling rule:** if (a, b, c) is a triple, so is (ka, kb, kc) for every positive integer k, because $(ka)^2 + (kb)^2 = k^2(a^2 + b^2) = k^2c^2 = (kc)^2$. Hence there are **infinitely many** triples.
- Every triple is a primitive triple or a scaled version of one.

▸ Generating triples from odd numbers

The sum of the first n odd numbers is n^2, so **$(n-1)^2 + (2n-1) = n^2$**. If the n-th odd number $2n-1$ is itself a perfect square, we get a triple.

Odd square $2n-1$	n	Triple
9	5	(3, 4, 5)
25	13	(5, 12, 13)
49	25	(7, 24, 25)
81	41	(9, 40, 41)
121	61	(11, 60, 61)

> **NOTE**
> This method always gives triples in which the hypotenuse is **one more** than a shorter side, so it does not give all primitive triples; for example (8, 15, 17) does not arise this way.

Fermat's Last Theorem

Since many squares are sums of two squares, Fermat asked about higher powers: **$x^n + y^n = z^n$** has **no solution in positive integers for $n > 2$**. This was only proved in **1994 by Andrew Wiles**, after more than 300 years.

Applications

▸ The lotus problem (Līlāvatī, Bhāskarāchārya)

A reed stands vertically with its tip **2 units** above the water. Bent over, the tip touches the water **6 units** away. Let the depth be x; the reed's length is $x + 2$ and we get a right triangle with sides 6, x, $x + 2$. Then $6^2 + x^2 = (x + 2)^2$, so $36 = 4x + 4$ and **$x = 8$**. The depth is 8 units.

Application	Method	Result
Diagonal of a square of side 9 cm	$d^2 = 2 \times 81$	$d = 9\sqrt{2}$ cm (≈ 12.7 cm)
Side of a rhombus with diagonals 10 and 24	The diagonals bisect each other at right angles: $(5)^2 + (12)^2 = \text{side}^2$	side = 13
Height of an equilateral triangle of side 8	The altitude bisects the base: $h^2 = 64 - 16 = 48$	$h = 4\sqrt{3}$, area $= \frac{1}{2} \times 8 \times 4\sqrt{3} = 16\sqrt{3}$
Right triangle with hypotenuse 25, one side 7	$b^2 = 625 - 49$	b = 24

Common Mistakes

- Using $a^2 + b^2 = c^2$ with the wrong side as c (the hypotenuse must be the longest side).
- Writing $(a + b)^2 = c^2$ instead of $a^2 + b^2 = c^2$.
- Applying the theorem to a triangle that is not right-angled.
- Stopping at c^2 and forgetting to take the square root.

Likely question	Model answer
What is the area of the square on the diagonal of a square of side 6?	Double the original area: 2 × 36 = 72 square units.
State the Baudhāyana–Pythagoras theorem.	In a right triangle, the square on the hypotenuse equals the sum of the squares on the other two sides: $a^2 + b^2 = c^2$.
Is (6, 8, 10) a primitive triple?	No, it is 2 × (3, 4, 5), a scaled version; 6, 8, 10 have the common factor 2.
Find the hypotenuse of a right triangle with sides 9 and 12.	$c^2 = 81 + 144 = 225$, so c = 15 (a scaled (3, 4, 5)).
Between which two decimals does √2 lie?	$1.414 < \sqrt{2} < 1.415$.

★ KEY TAKEAWAYS

- The square on the diagonal of a square has double its area; the side of the doubled square is $a\sqrt{2}$.
- √2 lies between 1.414 and 1.415 and is neither a terminating decimal nor a fraction of whole numbers.
- Baudhāyana–Pythagoras theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse of a right-angled triangle.
- The hypotenuse is the longest side; the theorem is used only for right-angled triangles.
- (ka, kb, kc) is a triple whenever (a, b, c) is, so there are infinitely many triples; primitive triples have no common factor.
- $(n - 1)^2 + (2n - 1) = n^2$ gives triples when 2n − 1 is a perfect square.
- Applications include the lotus problem, diagonals of rectangles and squares, rhombus sides and heights of equilateral triangles.